Polar Coordinates: Cartesian Equations

Questions 1–26

Question 1. A point has Cartesian coordinates (x, y) and polar coordinates (r, θ) where $r \geq 0$ and $-\pi < \theta \leq \pi$.

a) Express x and y in terms of r and θ.

b) Express r and θ in terms of x and y given $x > 0$ and $y > 0$.

Question 2. Find the Cartesian equation for each of the following curves:

a) $r = 3$

b) $r \sin\theta = 5$

c) $r \cos\theta = 2$

d) $r = 4(1 + \cos\theta)$

e) $r = 2(1 - \sin\theta)$

f) $r + r\cos\theta = 4$

g) $r = 4\cos\theta + 2\sin\theta$

h) $r = -2\cos\theta$

i) $r(3 + \sin\theta) = 1$

j) $r = \tan\theta$

Question 3. Find the polar equation for each of the following curves, where $r > 0$, writing your answer in the form $r = f(\theta)$:

a) $x^2 + y^2 = 25$
b) $x^2 + y^2 = 4x$
c) $xy = 9$
d) $y = x^2 - 3x$
e) $\frac{x^2}{16} + \frac{y^2}{9} = 1$
f) $(x^2 + y^2)^2 = x^2 - y^2$
g) $x^2 y = 6$

Question 4. A curve is defined by polar equation

$$r = 2\cos\theta - 5\sin\theta$$

where $0 < \theta \leq 2\pi$.

a) By converting the equation into its Cartesian form, show that the curve is a circle.
b) Hence state the coordinates of the centre of the circle in Cartesian form.
c) Find the radius of the circle.

Question 5. Show that the Cartesian equation

$$x^2 + y^2 + 4y = 12$$

may be written as a polar equation in the form

$$r = -a\sin\theta + \sqrt{b + c\sin^2\theta}$$

where $r > 0$ and a, b and c are positive constants to be found.

Question 6. Find the Cartesian equation of each of the curves, where $a > 1$:

a) $r = a\cos\theta$
b) $r = a\sin\theta$
c) $r = a(1 + \cos\theta)$
d) $r = a(1 - \sin\theta)$

Question 7. A circle is given by the polar equation

$$r = a(\cos\theta + \sin\theta)$$

where $r > 0$ and $0 < \theta \leq 2\pi$.

a) Find the radius of the circle in terms of a.

b) State the centre of the circle in Cartesian coordinates.

Question 8. Given that the polar equation of a curve is

$$r = a(p\cos\theta + q\sin\theta)$$

where $0 < \theta \leq 2\pi$ and a, p and q are positive constants,

a) By considering the Cartesian form of the curve, show that the curve is a circle.

b) Hence state the coordinates of the centre of the circle in Cartesian form.

c) Find the radius of the circle.

Prerequisite knowledge for Q9–Q11: Complex Loci

Question 9.

a) Show on an Argand diagram the locus of the points given by $|z - 8 + 6i| = 10$.

b) Show that this locus of points can be represented by the polar curve given by $r = 16\cos\theta - 12\sin\theta$.

Question 10.

a) Show on an Argand diagram the locus of the points given by $|z + 2 - 4i| = \sqrt{20}$.

b) Show that this locus of points can be represented by the polar curve given by $r = -4\cos\theta + 8\sin\theta$.

Question 11.

a) Show on an Argand diagram the locus of the points given by $|z - 4i| = |z + 10i|$.

b) Write the locus in the form $r = f(\theta)$.

Question 12. Write each of the following polar equations where $-\pi < \theta \leq \pi$ in the form

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

where a and b are positive integers to be found.

a) $r = \frac{1}{\sqrt{\cos^2\theta + \sin^2\theta}}$

b) $r = \frac{4}{\sqrt{16\cos^2\theta + \sin^2\theta}}$

c) $r = \frac{20}{\sqrt{25\cos^2\theta + 16\sin^2\theta}}$

d) $r = \frac{6}{\sqrt{9\cos^2\theta + 4\sin^2\theta}}$

Question 13. Write each of the following Cartesian equations where $-\pi < \theta \leq \pi$ in the form

$$r = \frac{a}{\sqrt{b\cos^2\theta + c\sin^2\theta}}$$

where $r > 0$ and a, b and c are positive integers to be found.

a) $\frac{x^2}{4} + \frac{y^2}{25} = 1$

b) $\frac{x^2}{16} + \frac{y^2}{9} = 1$

c) $\frac{9}{16}x^2 + \frac{49}{64}y^2 = 1$

Question 14. The curve C has polar equation

$$r = \frac{6}{4 + 2\sin\theta}$$

for $-\pi < \theta \leq \pi$.

a) Show that the maximum value of r is 3.

b) Show that the minimum value of r is 1.

c) Find the Cartesian equation of the curve C in the form $x^2 = f(y)$.

d) Hence, write the Cartesian equation in the form

$$\frac{x^2}{a} + \frac{(y+k)^2}{b} = 1$$

where a, b and k are integers to be found.

Question 15. The curve C has polar equation

$$r = \frac{16}{3 - \sin\theta}$$

for $-\pi < \theta \leq \pi$.

a) Show that the maximum value of r is 8.
b) Show that the minimum value of r is 4.
c) Find the Cartesian equation of the curve C in the form $x^2 = f(y)$.
d) Hence, write the Cartesian equation in the form

$$\frac{x^2}{a} + \frac{(y-k)^2}{b} = 1$$

where a, b and k are integers to be found.

Question 16. The curve C has polar equation

$$r = \frac{k}{2 + \cos\theta}$$

for $-\pi < \theta \leq \pi$. Write the equation of the curve C in the form

$$\frac{(x-a)^2}{b} + \frac{y^2}{c} = 1$$

where a, b and c are given in terms of k. **Prerequisite knowledge for Q17–Q19: Double Angle Formulae**

Question 17. Find the Cartesian equation for the curve given by the polar equation:

a) $r^2 \sin 2\theta = 9$

b) $r^2 \cos 2\theta = 4$

c) $r^4 \sin 4\theta = 4$

Question 18. Show that the polar equation $r^4 \cos 4\theta = 1$ has Cartesian form

$$x^4 - 6x^2y^2 + y^4 = 1$$

Question 19. The Cartesian equation of a curve is given by

$$x^4 + y^4 = x^2 + y^2$$

where $x \neq 0$ and $y \neq 0$.

a) Find the polar equation of the curve in the form

$$r^2 = \frac{p}{q - \sin^2 2\theta}$$

for $0 < \theta \leq 2\pi$ where p and q are constants to be found.

b) Find the polar coordinates of the points furthest away from the pole in the form (r, θ).

Question 20. The Cartesian equation of a curve is given by

$$x^3 + y^3 = 12xy$$

where $x > 0$ and $y > 0$.

a) Find a polar equation of the curve in the form $r = f(\theta)$.

b) Given that $0 < \theta \leq \frac{\pi}{2}$, find the values of θ when $r = 0$.

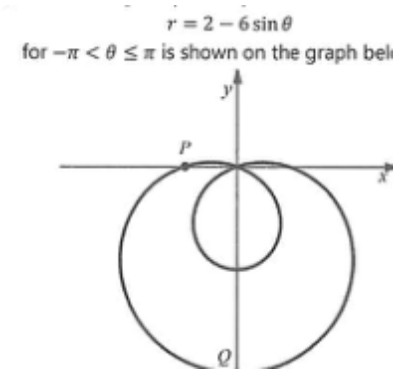
c) It is known that the curve has exactly one line of symmetry. Find the exact value of r when $\theta = \frac{\pi}{4}$.

d) Hence find the line of symmetry of the curve in its Cartesian form.

Question 21. The curve C_1 has polar equation

$$r = 2 - 6\sin\theta$$

for $-\pi < \theta \leq \pi$ and is shown on the graph below:



a) Find the Cartesian coordinates of point P.
b) State the polar coordinates of the point furthest away from the pole, Q.
c) A second curve, C_2, given by $x^2 + y^2 = 25$, intersects the curve C_1 at points M and N. Show that the area of triangle MNQ is given by $\frac{55\sqrt{3}}{4}$ square units.

Question 22. The curve C_1 has polar equation

$$r(3 + 2\cos\theta) = 1$$

a) Find the Cartesian equation of C_1 in the form $ay^2 = (1 - 5x)(1 + x)$ where a is a constant to be found.
The curve C_2 has Cartesian equation $9y^2 = x(6 - 5x)$.
b) Describe fully a single transformation that maps the curve C_1 onto the curve C_2.

Question 23. The curve C_1 with polar equation

$$r = \frac{7}{4 - 3\cos\theta}$$

is mapped onto the curve C_2 with Cartesian equation

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

by a single geometrical transformation.

a) By writing the curve C_1 as a Cartesian equation in a suitable form, find the values of a and b.

b) Hence state the geometrical transformation that maps curve C_1 onto curve C_2.

Question 24. The curve C_1 with polar equation

$$r = \cos\theta - \frac{1}{\sin\theta}$$

is mapped onto the curve C_2 with Cartesian equation

$$y^2x + x^3 = xy - x^2 - y^2$$

by a single geometrical transformation. Determine the geometrical transformation that maps curve C_1 onto curve C_2.

Question 25. A curve C_1 is described by Cartesian equation

$$(x^2 + y^2 - x)^2 = x^2 + y^2$$

and a second curve C_2 is described by polar equation

$$r = 1 - \sin\theta$$

for $0 \leq \theta \leq 2\pi$ and $r > 0$ respectively.

a) Find all the points of intersection of the two curves in exact Cartesian form.

b) Find the line of symmetry in its Cartesian form when both curves are plotted in the Cartesian plane.

Question 26. Two curves C_1 and C_2 are described by polar equations

$$r^2 = \cos\theta \quad \text{and} \quad r = \frac{52}{129 - 100\cos\theta}$$

for $0 \leq \theta \leq \frac{\pi}{2}$ and $r > 0$ respectively. The point N is the furthest point from the pole on curve C_2. The two curves intersect at points P and Q. Find the angle PNQ giving your answer in degrees to 3 significant figures.